

THE CHOICE OF ADDITIVE CONSTANTS IN RIDGE REGRESSION

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SUMMARY

Additive constants to be used in ridge regression are obtained by minimising estimated mean squared error functions.

1. Introduction

Hoerl and Kennard (1970a, b) proposed a ridge regression method for estimating regression weights. This method can give more precise estimates than the usual least squares procedure if certain additive constants are chosen in a suitable manner. Hoerl and Kennard (1970a, b) originally used a subjective choice procedure when a single additive constant was employed, and suggested an iterative procedure for obtaining a set of additive constants. Recently a number of other methods for choosing the additive constants have been proposed: Dempster (1973), Dempster, Schatzoff and Wermuth (1975), Farebrother (1975), Guilkey and Murphey (1975), Hemmerle (1975), Hoerl, Kennard and Baldwin (1975), McDonald and Galarneau (1975) and Obenchain (1975).

One principle which may be used for choosing additive constants is that of minimising an estimated mean squared error function (cf. Mallows (1973)). The present paper is concerned with the implementation of this principle. Two different estimated mean squared error functions are employed. One estimates the function used by Hoerl and Kennard (1970a); the other is a linear function of Mallows (1973) C_L statistic. The two approaches yield the same estimates when a set of additive constants is used but lead to slightly different solutions for the case of a single additive constant. Examples employed by Hoerl and Kennard (1970b) are used to illustrate the proposed methods for choosing additive constants.

2. Mean squared error criteria

We consider the usual regression model:

$$(2.1) \quad y = 1\mu + X\beta + e$$

where $\mathbf{1}$ represents the unit vector of order n , $\mathbf{X}$ is a known $n \times p$ matrix of rank p , $E(e) = 0$ and $E(ee') = \sigma^2 \mathbf{I}_n$. Without loss of generality we assume that $\mathbf{X}'\mathbf{1} = 0$.

The sample mean $\bar{y}$ is used to estimate μ . Let $S_{xx} = n^{-1} \mathbf{X}'\mathbf{X}$ and let $s_{xy} = n^{-1} \mathbf{X}'\mathbf{y}$. The ridge regression estimator of the vector of regression weights, β , may be expressed in the form (c.f. Hoerl and Kennard (1970a), p. 63):

$$(2.2) \quad \hat{\beta} = \mathbf{U}(\mathbf{D}_\lambda + \mathbf{D}_\kappa)^{-1} \mathbf{w},$$

where the orthogonal matrix $\mathbf{U}$ and diagonal matrix $\mathbf{D}_\lambda$ represent the eigenvectors and eigenvalues of S_{xx} and

$$\mathbf{w} = \mathbf{U}' s_{xy}.$$

The additive constants are represented by the diagonal matrix $\mathbf{D}_\kappa$ which may have equal diagonal elements.

A possible criterion for the effectiveness of the estimator $\hat{\beta}$, when $\mathbf{D}_\kappa$ is fixed, is the expected mean squared error of prediction,

$$(2.3) \quad c_1 = n^{-1} E_{y^*} E_y \{ (y^* - 1\bar{y} - \mathbf{X}\hat{\beta})' (y^* - 1\bar{y} - \mathbf{X}\hat{\beta}) \}.$$

Here y^* represents a future observation from (2.1) distributed independently of the observation y which is used to obtain $\bar{y}$ and $\hat{\beta}$.

Let

$$\mathbf{Z} = \mathbf{X}\mathbf{U},$$

$$\alpha = \mathbf{U}'\beta,$$

$$\hat{\alpha} = \mathbf{U}'\hat{\beta},$$

so that $\mathbf{Z}$ is the matrix of principal component regressors, α is the corresponding vector of population regression weights and $\hat{\alpha}$ is the ridge regression estimate of α with expected value

$$E_y \{ \hat{\alpha} \} = \mathbf{D}_\lambda (\mathbf{D}_\lambda + \mathbf{D}_\kappa)^{-1} \alpha$$

and covariance matrix

$$\text{Cov}_y (\hat{\alpha}, \hat{\alpha}') = n^{-1} \sigma^2 \mathbf{D}_\lambda (\mathbf{D}_\lambda + \mathbf{D}_\kappa)^{-2}.$$

Then expression of (2.3) in the equivalent form

$$c_1 = n^{-1} E_{y^*} E_y \{ (y^* - 1\bar{y} - \mathbf{Z}\hat{\alpha})' (y^* - 1\bar{y} - \mathbf{Z}\hat{\alpha}) \},$$

and evaluation of the expectations shows that

$$(2.4) \quad c_1 = n^{-1} \sigma^2 [n + 1 + \text{tr} \{D_\lambda^2 (D_\lambda + D_K)^{-2}\}] + \alpha' D_\lambda D_K^2 (D_\lambda + D_K)^{-2} \alpha.$$

Thus c_1 is a function of the unknown population parameters σ^2 and α . Given a fixed D_K , an unbiased estimator of c_1 is

$$\begin{aligned} \hat{c}_1 &= n^{-1} \hat{\sigma}^2 [n + 1 + \text{tr} \{(D_\lambda - D_K) (D_\lambda + D_K)^{-1}\}] \\ &\quad + w' D_K^2 D_\lambda^{-1} (D_\lambda + D_K)^{-2} w \\ (2.5) \quad &= v^2 + 2n^{-1} \hat{\sigma}^2 [1 + \text{tr} \{D_\lambda (D_\lambda + D_K)^{-1}\}] \end{aligned}$$

where

$$\begin{aligned} v^2 &= n^{-1} (y - 1\bar{y} - X\hat{\beta})'(y - 1\bar{y} - X\hat{\beta}) \\ &= s_y^2 - w'(D_\lambda + 2D_K) (D_\lambda + D_K)^{-2} w, \\ \hat{\sigma}^2 &= n(n-p-1)^{-1} (s_y^2 - s_{xy}' S_{xx}^{-1} s_{xy}) \\ &= n(n-p-1)^{-1} (s_y^2 - w' D_\lambda^{-1} w) \end{aligned}$$

and

$$s_y^2 = n^{-1} y'y - \bar{y}^2.$$

Here v^2 represents the average squared residual and $\hat{\sigma}^2$ is the usual unbiased estimator of σ^2 .

The statistic $\hat{c}_1$ is linearly related to Mallows (1973) C_L statistic:

$$(2.6) \quad \hat{c}_1 = \hat{\sigma}^2 (n^{-1} C_L + 1).$$

Another mean square error criterion for the effectiveness of $\hat{\beta}$ was proposed by Hoerl and Kennard (1970a),

$$\begin{aligned} c_2 &= E \{(\hat{\beta} - \beta)' (\hat{\beta} - \beta)\} \\ (2.7) \quad &= n^{-1} \sigma^2 \text{tr} \{D_\lambda (D_\lambda + D_K)^{-2}\} + \alpha' D_K^2 (D_\lambda + D_K)^{-2} \alpha \end{aligned}$$

An unbiased estimator of c_2 , given a fixed D_k , is

$$\hat{c}_2 = n^{-1} \hat{\sigma}^2 \text{tr} \{ (D_\lambda - D_k) D_\lambda^{-1} (D_\lambda + D_k)^{-1} \} \\ + w' D_k^2 D_\lambda^{-2} (D_\lambda + D_k)^{-2} w.$$

3. Estimation of D_k when the diagonal elements differ

Minimisation of c_1 or c_2 with respect to D_k would give a solution for the additive constants which involves population parameters. We shall therefore minimise $\hat{c}_1$ or $\hat{c}_2$ with respect to D_k , *subject to the inequality constraints* $\kappa_i \geq 0$, $i = 1, \dots, p$.

Let

$$(3.1) \quad t_i = w_i^2 - n^{-1} \hat{\sigma}^2 \lambda_i, \quad i = 1, \dots, p.$$

The first two partial derivatives of $\hat{c}_1$ and $\hat{c}_2$ with respect to the additive constants are:

$$\frac{\partial \hat{c}_1}{\partial \kappa_i} = \frac{2(t_i \kappa_i - n^{-1} \hat{\sigma}^2 \lambda_i^2)}{(\lambda_i + \kappa_i)^3}$$

$$\frac{\partial^2 \hat{c}_1}{\partial \kappa_i^2} = \frac{2(w_i^2 \lambda_i + 2n^{-1} \hat{\sigma}^2 \lambda_i^2 - 2t_i \kappa_i)}{(\lambda_i + \kappa_i)^4}$$

$$\frac{\partial \hat{c}_2}{\partial \kappa_i} = \frac{1}{\lambda_i} \frac{\partial \hat{c}_1}{\partial \kappa_i},$$

$$\frac{\partial^2 \hat{c}_2}{\partial \kappa_i^2} = \frac{1}{\lambda_i} \frac{\partial^2 \hat{c}_1}{\partial \kappa_i^2}.$$

Examination of these derivatives will show that the required minima of both $\hat{c}_1$ and $\hat{c}_2$ are given by:

$$(3.2) \quad \hat{\kappa}_i = \frac{n^{-1} \hat{\sigma}^2 \lambda_i^2}{t_i} \text{ if } t_i > 0$$

and

$$(3.3) \quad \hat{\kappa}_i = +\infty \text{ if } t_i \leq 0$$

It follows from (2.2), (3.2) and (3.3) that the ridge regression estimate of β is given by

$$(3.4) \quad \hat{\beta} = U\hat{\alpha}$$

where

$$\hat{\alpha}_i = \frac{w_i}{\lambda_i + \hat{\kappa}_i}, \quad i = 1, \dots, p,$$

so that

$$(3.5) \quad \hat{\alpha}_i = \frac{t_i}{w_i \lambda_i} \text{ if } t_i > 0,$$

and

$$(3.6) \quad \hat{\alpha}_i = 0 \text{ if } t_i \leq 0.$$

It was pointed out earlier that $\hat{\alpha}$ is the ridge regression estimate for the regression weight vector, α , of the regressand, y , on the principal component regressors Z . Thus minimising $\hat{c}_1$ or $\hat{c}_2$ subject to the constraints that the additive constants, $\hat{\kappa}_i$, be nonnegative results in zero estimated regression weights, $\hat{\alpha}_i$, for those principal component regressors whose covariances, w_i , with the regressand are "small" relative to their variances, λ_i .

Examination of (3.2) shows that the additive constants $\hat{\kappa}_i$ can be expected to decrease as the sample size n increases. It can be seen from (3.1) and (3.6), also, that principal component regressors are less likely to be discarded when n is increased.

Hoerl and Kennard (1970a, Section 5) minimised c_2 with respect to D_K to obtain

$$(3.7) \quad \kappa_i = \frac{n-1}{\alpha_i^2} \sigma^2.$$

It is of interest to note that replacement of the numerator and denominator of (3.7) by unbiased estimators gives (3.2) which, with (3.3), is equivalent to Obenchain's (1975) equation (5.2).

4. Estimation of κ when $D_K = \kappa I$

When a single additive constant is to be employed, minimisation of $\hat{c}_1$ is no longer equivalent to minimisation of $\hat{c}_2$. Furthermore the solution for the additive constant which minimises $\hat{c}_1$ or $\hat{c}_2$ can no longer be expressed in closed form. This additive con-

stant can, however, be obtained iteratively by means of Newton's method.

It is convenient to express $\hat{c}_1$ and $\hat{c}_2$ in the common form

$$(4.1) \quad \hat{c}_\ell(\kappa) = (2-\ell)(n+1)n^{-1}\hat{\sigma}^2 + \sum_{i=1}^p (n^{-1}\hat{\sigma}^2\lambda_i^3 + t_i\kappa^2) / [\{2-\ell+(\ell-1)\lambda_i\} \lambda_i(\lambda_i+\kappa)^2]$$

where $\ell = 1$ for $\hat{c}_1$ and $\ell = 2$ for $\hat{c}_2$. We require the value $\hat{\kappa}$ of κ which yields the absolute minimum of $\hat{c}_\ell(\kappa)$ over the region $\kappa \geq 0$. When $\ell = 1$ this $\hat{\kappa}$ will also minimise Mallows C_L (cf. 2.6).

Let

$$\begin{aligned} g_\ell(\kappa) &= -\frac{1}{2} \partial \hat{c}_\ell(\kappa) / \partial \kappa \\ &= \sum_{i=1}^p (n^{-1}\hat{\sigma}^2\lambda_i^2 - t_i\kappa) / [\{2-\ell+(\ell-1)\lambda_i\} (\lambda_i+\kappa)^3] \end{aligned}$$

and

$$\begin{aligned} h_\ell(\kappa) &= \frac{1}{2} \partial^2 \hat{c}_\ell(\kappa) / \partial \kappa^2 \\ &= \sum_{i=1}^p \{ w_i^2 \lambda_i + 2(n^{-1}\hat{\sigma}^2\lambda_i^2 - t_i\kappa) \} / [\{2-\ell+(\ell-1)\lambda_i\} (\lambda_i+\kappa)^4]. \end{aligned}$$

The Newton algorithm then is:

$$(4.2) \quad \hat{\kappa}_{(j+1)} = \hat{\kappa}_{(j)} + g_\ell(\hat{\kappa}_{(j)}) / h_\ell(\hat{\kappa}_{(j)}).$$

where $\hat{\kappa}_{(j)}$ represents the j^{th} approximation for $\hat{\kappa}$.

In some instances $\hat{c}_\ell(\kappa)$ will have more than one minimum over the region $\kappa \geq 0$ and the final solution yielded by the Newton algorithm depends on the starting point, $\hat{\kappa}_{(0)}$. In an attempt to find a starting point near the absolute minimum we take $\hat{\kappa}_{(0)}$ to be equal to that $\hat{\kappa}_i$ defined by (3.2) and (3.3) which, when substituted for κ in (4.1), yields the smallest value of $\hat{c}_\ell(\hat{\kappa}_i)$, provided that the second derivative, $h_\ell(\hat{\kappa}_i)$, is positive at this point. If $h_\ell(\hat{\kappa}_i)$ is negative, the next largest $\hat{\kappa}_i$ is chosen and this procedure is continued until a $\hat{\kappa}_i$ which gives a positive value of $h_\ell(\hat{\kappa}_i)$ is found. Such a $\hat{\kappa}_i$ can always be found since $h_\ell(\hat{\kappa}_i)$ is positive at the smallest $\hat{\kappa}_i$.

If the starting point $\hat{\kappa}_{(0)}$ chosen in this manner is infinite (cf. 3.3) no iteration is carried out and the estimate of β is taken to be

$$\hat{\beta} = 0.$$

If $\hat{\kappa}_{(0)}$ is finite (4.2) is applied iteratively until convergence occurs. (In thirty or so applications of the procedure to artificially generated data convergence has been satisfactory). After convergence of (4.2) to $\hat{\kappa}$, the estimate of β is obtained from

$$\hat{\beta} = U(D_{\lambda} + \hat{\kappa}I)^{-1} w.$$

5. Examples

Two examples were used by Hoerl and Kennard (1970b); a ten factor regression problem of Gorman and Toman (1966) and a thirteen factor problem of Jeffers (1967). Additive constants for these two examples were obtained using the methods of Section 3 and Section 4. The correlation matrices given by Gorman and Toman (1966) and by Jeffers (1967) were employed. These differ slightly from those given by Hoerl and Kennard (1970b).

In both these examples $\hat{c}_1$ and $\hat{c}_2$ had single minima and, in the applications of Newton's method, convergence of the $\hat{\kappa}_{(j)}$ to six decimal places occurred cleanly in at most four iterations.

The additive constants chosen using present methods and the corresponding regression weights are given in Tables 1 and 2.

TABLE 1
Regression weights and additive constants
Gorman and Toman's example. $n = 36$.

i	$\hat{c}_1$ and $\hat{c}_2$ minimised $\hat{\kappa}_i$ different		$\hat{c}_1$ minimised $\hat{\kappa} = 0,0161$	$\hat{c}_2$ minimised $\hat{\kappa} = 0,0087$	Least Squares $\hat{\kappa} = 0$	Hoerl & Kennard $\hat{\kappa} = 0,25$
	$\hat{\kappa}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$
1	0,0297	-0,148	-0,229	-0,212	-0,185	-0,296
2	0,0404	-0,162	-0,200	-0,209	-0,221	-0,110
3	∞	-0,312	-0,346	-0,352	-0,359	-0,244
4	0,0894	-0,061	-0,096	-0,100	-0,105	-0,054
5	12,7230	-0,348	-0,379	-0,416	-0,469	-0,043
6	0,0623	0,785	0,693	0,742	0,813	0,332
7	0,0643	0,255	0,240	0,258	0,285	0,055
8	0,0497	0,369	0,366	0,373	0,383	0,244
9	0,0851	0,091	0,104	0,099	0,092	0,123
10	0,0062	0,070	0,105	0,101	0,094	0,125

TABLE 2
Regression weights and additive constants
Jeffers' example. $n = 180$

i	$\hat{c}_1$ and $\hat{c}_2$ minimised $\hat{\kappa}_i$ different		$\hat{c}_1$ minimised $\hat{\kappa} = 0,0063$	$\hat{c}_2$ minimised $\hat{\kappa} = 0,0048$	Least Squares $\hat{\kappa} = 0$	Hoerl & Kennard $\hat{\kappa} = 0,3$
	$\hat{\kappa}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$	$\hat{\beta}_i$
1	0,0925	-0,483	-0,435	-0,446	-0,488	-0,135
2	0,0116	0,366	0,343	0,355	0,401	0,000
3	0,1053	-0,934	-0,900	-0,915	-0,975	-0,406
4	1,9336	0,272	0,214	0,230	0,292	-0,187
5	0,0109	-0,084	-0,055	-0,060	-0,082	0,089
6	0,0221	0,126	0,182	0,181	0,179	0,150
7	0,0304	0,220	0,139	0,138	0,135	0,131
8	0,0639	-0,221	-0,267	-0,266	-0,263	-0,203
9	∞	-0,070	-0,067	-0,068	-0,073	-0,043
10	∞	-0,132	-0,080	-0,081	-0,085	-0,046
11	∞	0,107	0,095	0,096	0,097	0,055
12	0,0015	-0,066	-0,067	-0,068	-0,069	-0,042
13	∞	0,008	0,003	0,002	0,001	-0,006

In common with the additive constants yielded by other recently proposed methods (cf. Obenchain, 1975, Table 3) they are small. Values of the average squared residual, v^2 , and the estimated mean squared errors, $\hat{c}_1$ and $\hat{c}_2$, for each choice of the additive constants are shown in Tables 3 and 4.

TABLE 3
Estimated mean squared errors
Gorman and Toman's example

	v^2	$\hat{c}_1$	$\hat{c}_2$
$\hat{\kappa}_i$ different $\hat{c}_1$ and $\hat{c}_2$ minimised	0,1127	0,1796	0,1096
Single $\hat{\kappa}$			
$\hat{c}_1$ minimised	0,1054	0,1923	0,1349
$\hat{c}_2$ minimised	0,1040	0,1926	0,1304
$\hat{\kappa} = 0$ (Least Squares)	0,1033	0,1941	0,1397
$\hat{\kappa} = ,25$ (H. & K.)	0,1601	0,2230	0,5000

TABLE 4
Estimated, mean squared errors
Jeffers' example

	v^2	ϵ_1	ϵ_2
$\hat{\kappa}$ different $\hat{\epsilon}_1$ and $\hat{\epsilon}_2$ minimised	0,2717	0,3010	-0,0230
Single $\hat{\kappa}$			
$\hat{\epsilon}_1$ minimised	0,2699	0,3138	0,1288
$\hat{\epsilon}_2$ minimised	0,2696	0,3138	0,1280
$\kappa = 0$ (Least Squares)	0,2691	0,3145	0,1398
$\hat{\kappa} = 0,3$ (H. & K.)	0,3285	0,3561	0,7969

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